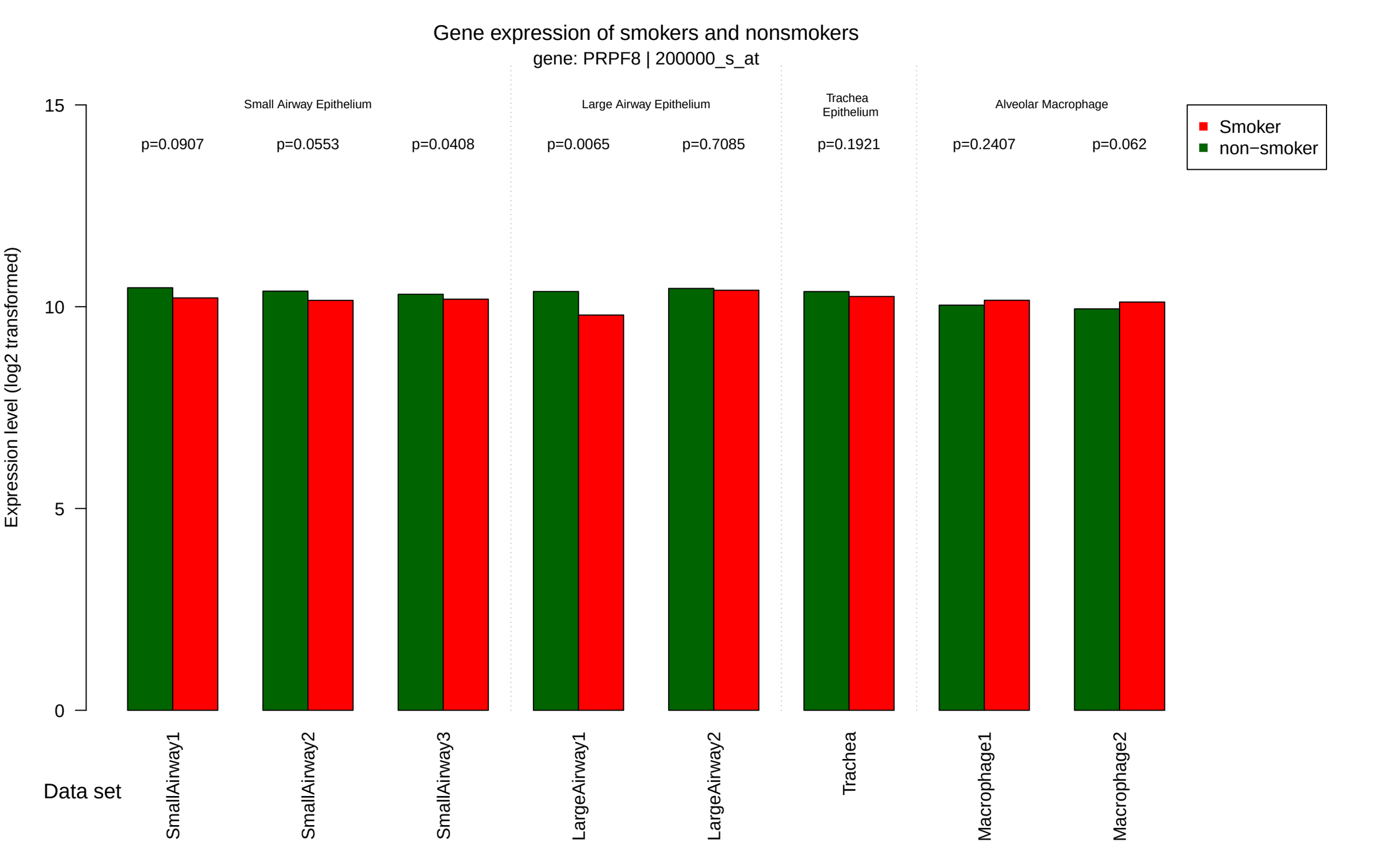
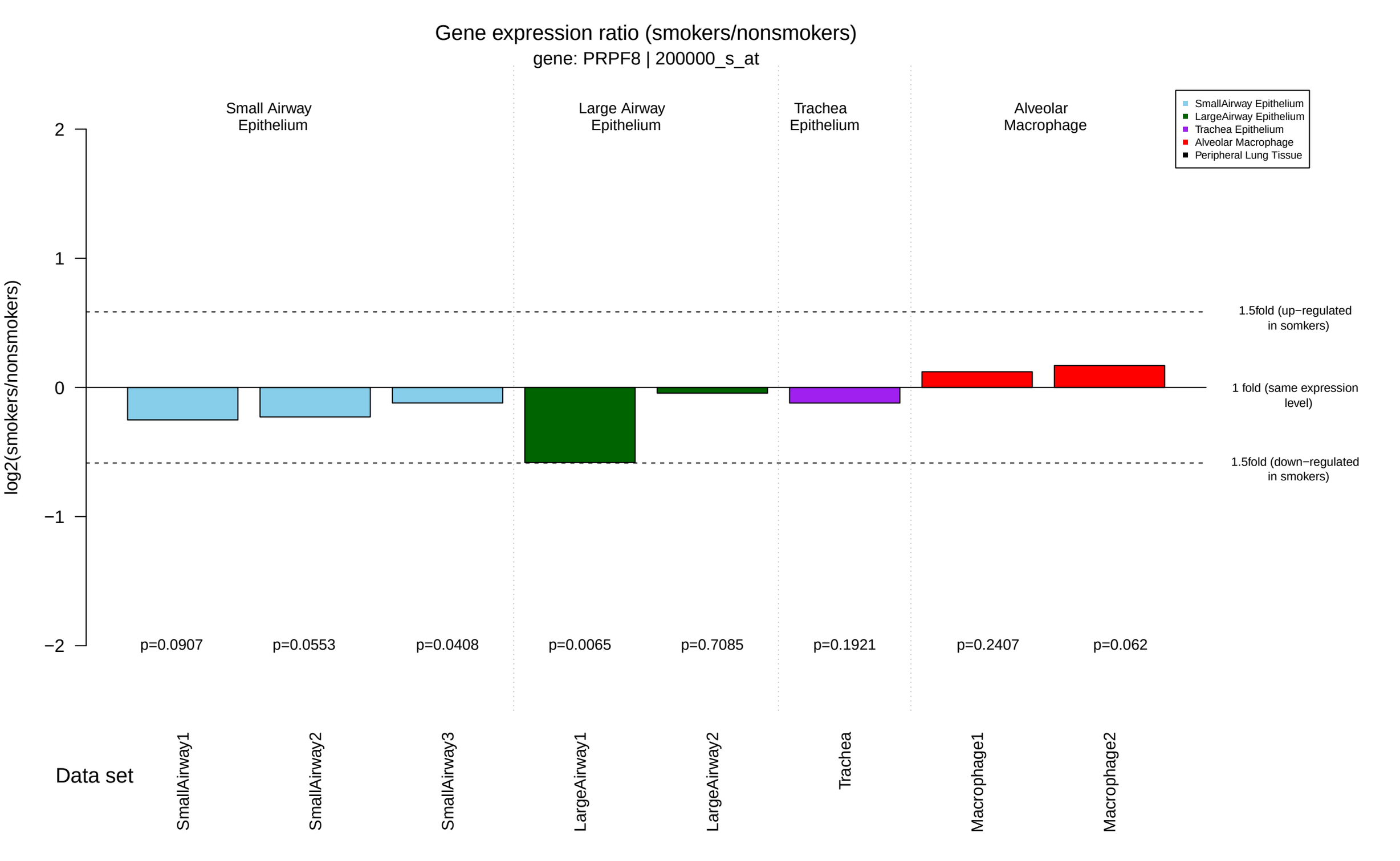
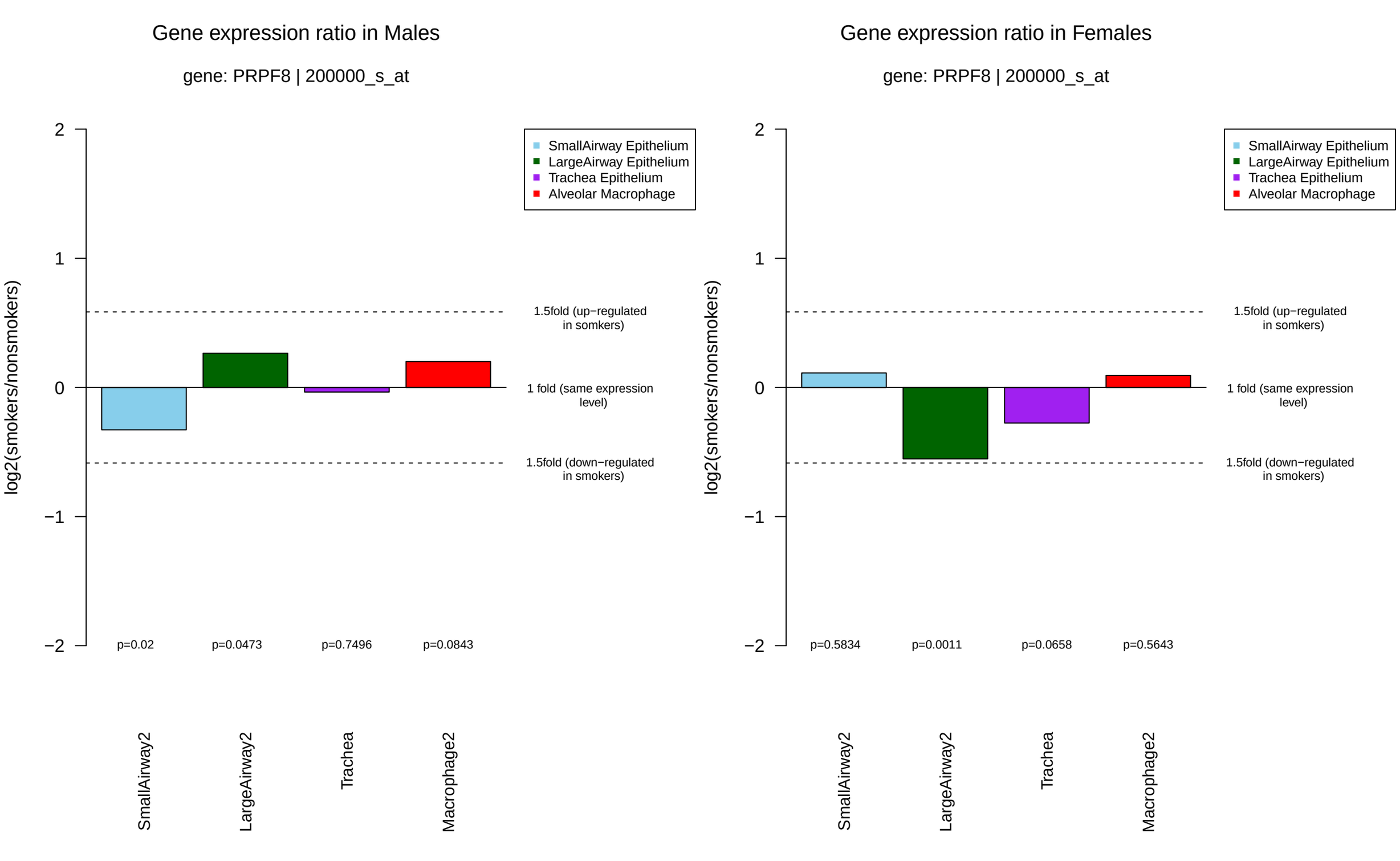
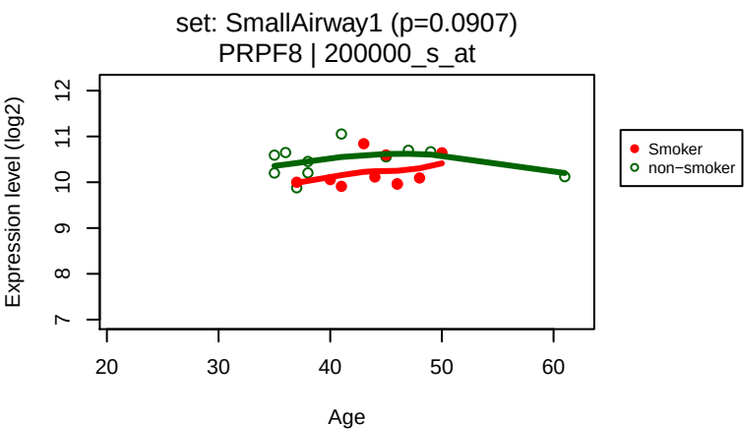
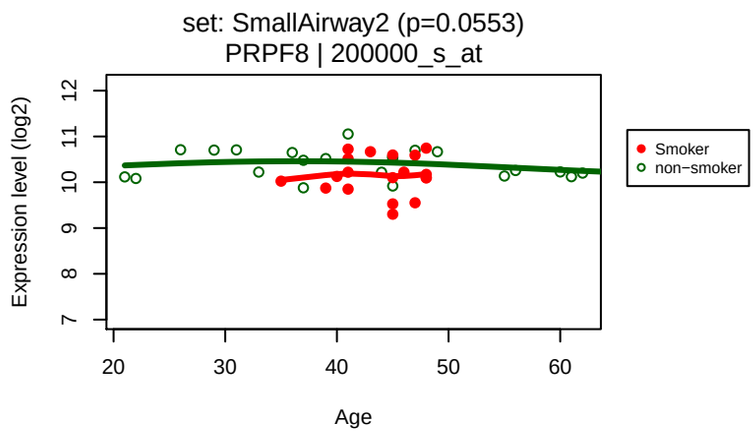
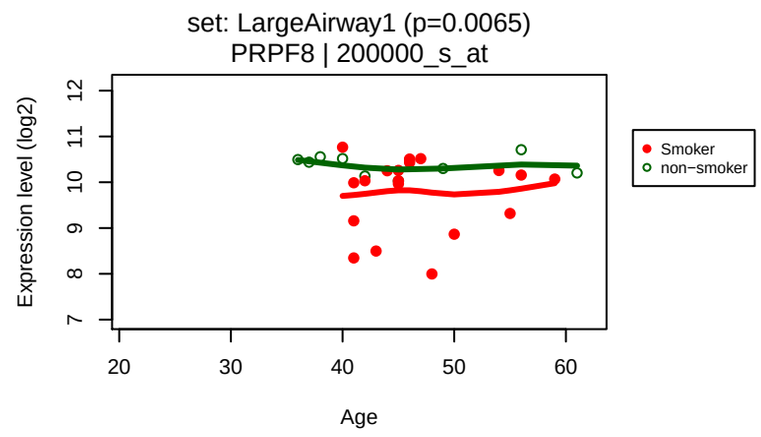

Gene expression of smokers and nonsmokers
gene: PRPF8 | 200000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0907
p=0.0553
p=0.0408
p=0.0065
p=0.7085
p=0.1921
p=0.2407
p=0.062
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

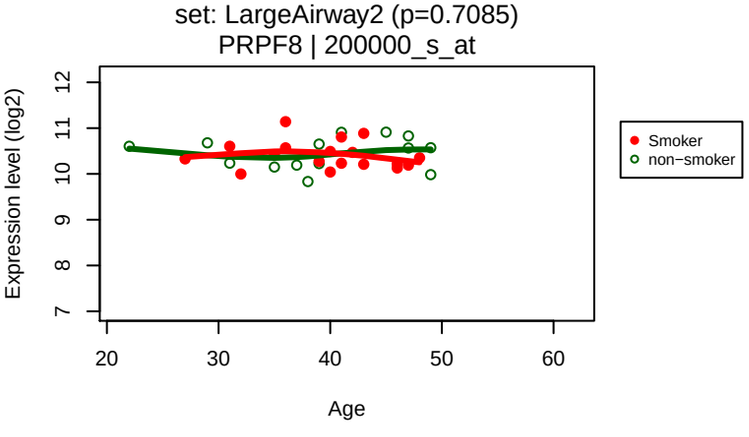

Gene expression ratio (smokers/nonsmokers)
gene: PRPF8 | 200000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0907
p=0.0553
p=0.0408
p=0.0065
p=0.7085
p=0.1921
p=0.2407
p=0.062
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

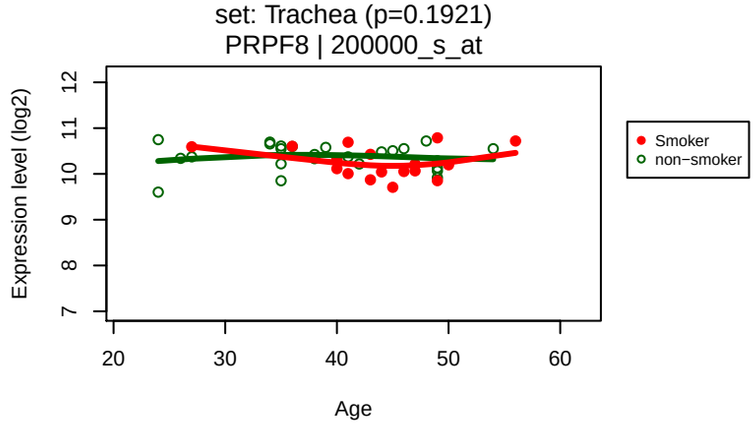

Gene expression ratio in Males
gene: PRPF8 | 200000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.02
p=0.0473
p=0.7496
p=0.0843
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRPF8 | 200000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5834
p=0.0011
p=0.0658
p=0.5643
SmallAirway2
LargeAirway2
Trachea
Macrophage2

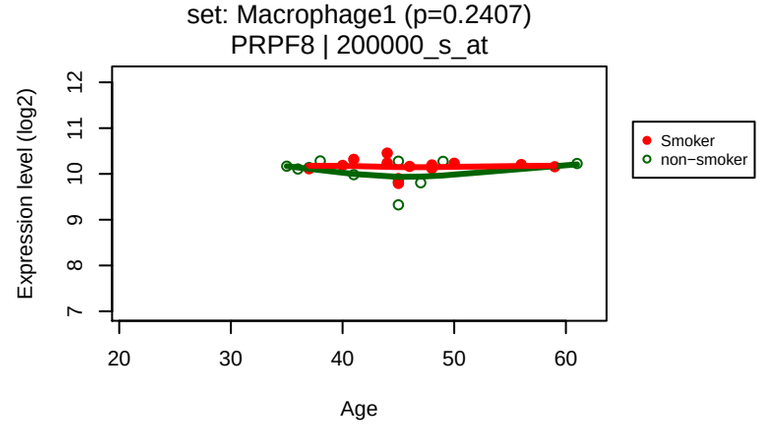

set: SmallAirway1 (p=0.0907)
PRPF8 | 200000_s_at
Expression level (log2)
Age
Smoker
non−smoker

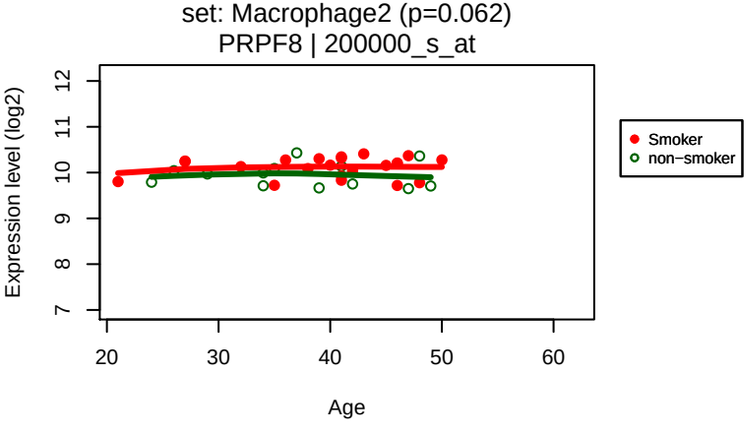

set: SmallAirway2 (p=0.0553)
PRPF8 | 200000_s_at
Expression level (log2)
Age
Smoker
non−smoker

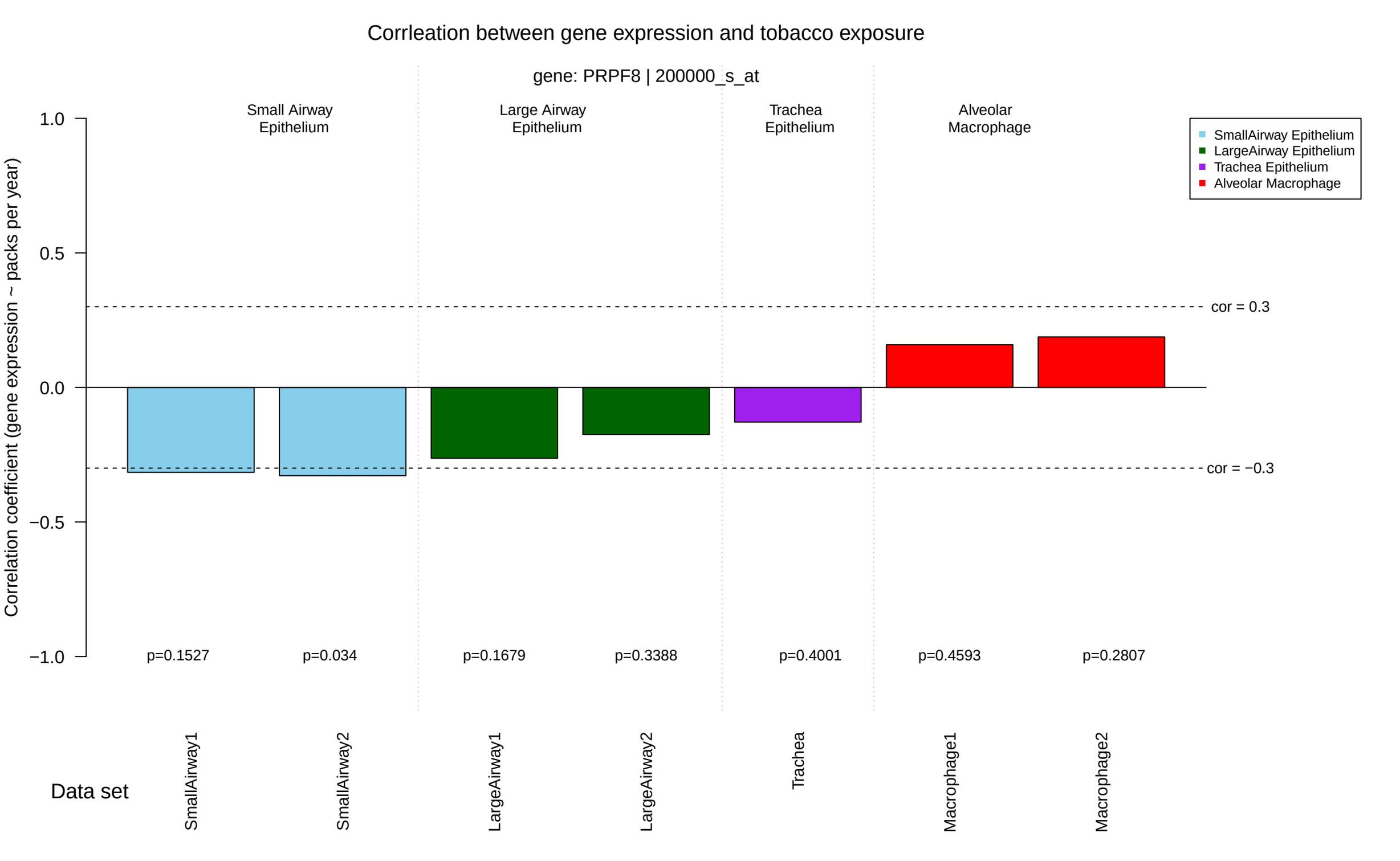

set: LargeAirway1 (p=0.0065)
PRPF8 | 200000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7085)
PRPF8 | 200000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1921)
PRPF8 | 200000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2407)
PRPF8 | 200000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.062)
PRPF8 | 200000_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRPF8 | 200000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1527
p=0.034
p=0.1679
p=0.3388
p=0.4001
p=0.4593
p=0.2807
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2